

Commercial Auto Risk Scoring Case Study

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1. Exploratory data analysis I.

Unclean data

- Placeholder and unclean data, wrong dates
- --> need to cleanup the data, normalize the values
- Business type was not consistent,
- --> different combinations if upper and lower case

Observations

- Trucking has the highest claim rate of 13.4% next to the construction - 12.9%
- Historical claims and losses are very important signal for the future claims
- It is strange that all claims were collected in cca 45 days (median 45, max 70)

After running the provided prototype:

- Claim reporting delays were surprisingly concentrated around 45 days (median = 45, maximum = 70).

2. Modeling methodology improvements I.

Data Quality Improvement - **input data issues to improve the model and calculation**

Normalization

- As described before – normalization of the business types and
- calls without values were filled with zeros -> now are filled with median value, which is better for training

Introduced **Scaling** for for a Logistic Regression model

- Introduced StandardScaler to standardize numeric inputs, and ensure reliable model training

2. Modeling methodology improvements II.

Methodology improvements

1. Eliminate Data Leakage

- Cannot train model on the data which will be then not yet available during the real usage of the systems

2. Dataset Split: Random Split -> Time-Based Validation

- This is not suitable for time series, here is better....
- Training – 2020 + 2021, validation in 2022

3. Evaluation methodology: Accuracy -> ROC/AUC

- Accuracy replaced with standard ROC/ AUC + PR/AUC - standard metrics used in the insurance business, explaining better the model capabilities
- ROC above 0.5 is actual prediction / from above 0.7 is model is usable
- PR used if there are not enough positive cases – like in the provided dataset

Accuracy = 0.874 ROC-AUC = 0.684 PR-AUC = 0.238

2. Modeling methodology improvements II.

Using LogisticRegression model -> Tree-based algorithm

- LogisticRegression. It is a good baseline, it is simple and fast, very good for prototyping, calculates linearly.
- For real usage shall be used some more complex like tree-based algorithms like RandomForestClassifier - captures better the real world with non-linear dependencies during considering multiple parameters
- Experimenting with the more advanced models (see the code) have not provided better results (ROC/AUC) as the sample data does not provide enough data to train the model, also there is not enough variables for the advanced model. Therefore, the Logistic Regression model was retained.

Used columns / features to train the model

- Some of the reasonable columns or real parameters shall be used to have more real results and these are not used - like safety program or payment frequency. These may also provide signals about the behavior
- Some columns shall be removed (data leakage)

Risk Score calculation

- Original prototype: -> predict() -> binary 0/1 output 4
- Improved solution: --> predict_proba() -> probability-based risk score
- Automation - low risk – no approval, high risk – special approval necessary

3. Production readiness improvements. I.

Input data validation

- Deeper validation of the input training data – removing the extreme and defected values (duplicates, mandatory params...)
- For production usage needed even more input data validation and normalization

For the QA testing before deployed to production

- Using reproducible training to be able to run the tests and production system repeatable
- Using the Fixed Random State to fix the training

In the category of production readiness is also the

- Usage of tree-based model discussed before which is more advanced
- Selection of more and reasonable features for better learning

4. Production readiness roadmap. I.

Scale to real size

- not using the csv but connect to database or Databricks, Azure

Monitoring

- Monitoring of the model in the real operation to verify the changing inputs – e.g. new vehicle types or ratios between cars and trucks..
- data drift and prediction drift- if the population in the training data is changing
- Score distribution – if the results start to shift

Periodical model update and automatic re-training

- Training of the model after reasonable period – quarter of half a year to eliminate model drift, and train the model on the recent claims experience
- Shifting the training data e.g. for 2025, start to learn on the data from Q4 2025

For the real deployment

- serious build and release pipelines and versioning of the models
- Using serious CI/CD pipelines

New concepts

Insurance-Oriented Evaluation Metrics

- Misleading accuracy
- ROC-AUC and PR-AUC

Probability-Based Risk Scoring

- difference between binary classification outputs (`predict()`) and probability-based scores (`predict_proba()`).
- For underwriting, probability scores are much more useful

Data Quality Challenges

- inconsistent categorical values, placeholder values, missing values can significantly affect model quality

Production requirements

reproducibility, model monitoring, drift detection deployment pipelines

AI tools

- AI tools used in working at this study case were **Copilot** to help me find more patterns, inconsistencies that were found by me. To explain the prototype python file and some financial word definitions and check the completion of all the tasks given.
- **Claude code** was used to fix the code bugs after the copilot as it is the better tool for this.
- **GitHub Copilot** to write comments in the code